SEP 2015 NUMERICAL METHOD 80 Marks

SECTION -A

Answer any ten Questions : **10*2=20**

1. Define floating point binary.
2. Define normalization.
3. Write the difference between Algebraic and Transcendental equation.
4. Write the formula to find the root of an equation using Secant Method.
5. For which system of equations Gauss Seidal method is applicable.
6. Differentiate between Direct and Iterative method.
7. Define Eigen value and Eigen vector of square matrix.
8. Define Extrapolation.
9. Define Shift operator 'E'.
10. Write the formula for Simpson's 3/8th rule for numerical integration.
11. What is curve fitting?
12. Define numerical differentiation.

SECTION -B

Answer any six Questions : **6*5=30**

13. Explain representation of a binary number in normalized floating point node with an example.
14. Using Bisection Method find the root of an equation $x^3 - 4x - 9 = 0$ that lies between 2 and carry out 3 iterations.
15. Explain the Newton Raphson Method to find the roots of an equation.
16. Solve by Gauss elimination Method:

$$2x + y + 10$$
$$3x + 2y + 3z = 18$$
$$x + 4y + 9z = 16$$

17. Fit a Straight line $y = a + bx$ in the least square sense for the data.

X	5	10	15	20	25
Y	16	19	23	26	30

18. Find the interpolating polynomial for the following data using Lagrange's Formula.

X	1	3	4
f(x)	1	27	64

19. From the table of values below compute $\frac{dy}{dx}$ at x=1.

X	1	2	3	4	5	6
Y	1	8	27	64	125	276

20. Using suitable interpolation formula find f(38).

X	40	50	60	70	80	90
Y=f(x)	184	204	226	250	276	304

SECTION - C

Answer any three Questions. **3*10=30**

21.
   (a) Determine the single precision machine representation of the decimal number -52.234275.
   (b) Explain Gauss Jordan method to solve system of linear simultaneous equation.

22.

(a) Use the method of false position to find a real root of the equation $x^3 - 4x - 5 = 0$ corrected to three decimal places.

(b) Find the real root of the equation $x^3 - 5x - 1 = 0$ using Secant method corrected to three decimal places.

23. Using LU decomposition method solve the system of equations

$$2x + y + 4z = 12$$
$$4x + 11y - z = 33$$
$$8x - 3y + 2z = 20$$

24. Using suitable interpolation formula find f(142) and f(175).

X	140	150	160	170	180
Y	3.685	4.854	6.302	8.076	10.225

25. Calculate $I = \int_0^1 dx \frac{dx}{1+x^2}$ dividing the interval [0.1] into six equal parts by using Trapezoidal and simpson's 1/3rd rule.

SEP 2015 STATISTICS 80 MARKS

SECTION –A

1. Answer any ten Questions of the following:
(a) What is primary and Secondary data?
(b) Define ratio.
(c) List applications of Statistics in business.
(d) Mention different types of classification.
(e) What is frequency distribution?
(f) Define Median.
(g) Define Standard Deviation.
(h) Define Quartile Deviation.
(i) What is Correlation? State its importance.
(j) What is Scatter diagram?
(k) What is cyclical variation?
(l) What is secular trend?

SECTION –B

Answer any four questions of the following: 5*4= 20

2.
(a) Define Proportion.
(b) If the cost of 15 Kg of Sugar is 12, find the cost of 25 kg of Sugar. (2+3)

3. From a Frequency distribution from the following data by inclusive method, taking 4 as magnitude of class interval.

10	17	15	22	11	16	19	24	29	18	25	26	32	14
17	20	23	27	30	12	15	18	24	36	18	15	21	28

4. Calculate standard deviation from the following series.
Wages in Rs : 40, 44, 54, 60, 62, 64, 70, 80, 90, 96.
5. Discuss the merits and demerits of a mode.
6. Given the following paris of the variables X and Y :

X	4	6	10	12	16	18
Y	12	10	14	16	24	22

Draw Scatter diagram and indicate whether the correlation id positive or negative.
7. State and Explain method of Semi averages in case of even number series.

SECTION –C

Answer any four questions of the following: 4*10=40

8.
(a) Write a note on the methods of data collection. 4m
(b) Explain the following data types: 6m
(i) Dichotomous
(ii) Continuous
(iii) Nominal

9.
(a) What are the main objectives of tabulation? 5m
(b) The following data relating to the strength of the Indian Merchant Shipping fleet gives the gross Registered Tonnage (GRT) as on 31st December, for different years. 5m

Year	1961	1966	1971	1975	1976
GRT in '000 :	901	1792	2500	4464	5115

Represent the data by a suitable bar diagram.

10.

(a) What are the chief requisites of a good measure of dispersion?

(b) Find the harmonic mean from the following data(Discrete Series):

X	10	20	40	60	120
y	01	3	6	5	4

11.

(a) State Karl Pearson's co-efficient of correlation, give its formula.

(b) Calculate Karl Pearson's co-efficient of correlation from the following data:

X	12	9	8	10	11	13	7
Y	14	8	6	9	11	12	3

12.

(a) What are the uses of a trend?

(b) Fit a trend line to the following data by the method of semi averages:

Year :	2006	2007	2008	2009	2010	2011	2012
Sales (in thousands)	51	54	57	55	54	58	56

SEP 2016 STATISTICS 80 MARKS

SECTION –A 10*2=20

1. Answer any ten of the following Questions:

(a) Define Statistics and explain its applications.
(b) Explain any two features of Statistics.
(c) Explain with example discrete and continuous variables.
(d) Define geographical classification with an example.
(e) Define correction factor and state how it is calculated.
(f) Explain any four parts of Table.
(g) State general rules for constructing diagrams and graphs.
(h) Define Arithmetic Mean and list its types.
(i) Define percentiles and explain steps of how it is accessed.
(j) Mention different types of correlation.
(k) Define Regression.
(l) Define Probability with examples.

SECTION – B

Answer any four of the following: 4*5=40 4*5=20

2. Following is the data regarding the I.Q. of 30 Children of a school:

106	118	112	120	103	105	99	139	65	113	108	116	93	100	120
100	108	100	112	110	109	117	98	103	99	110	105	101	105	98

Prepare a frequency distribution with class intervals : less than or equal to 90, 90-110, 110-120, more then or equal to 120.

3. In the sample study regarding smoking habit in a town, the following data were obtained :
Smokers = 22%
Men smokers = 18%
Tabulate the above data.
4. Explain method of construction of Bi-variate Frequency Distribution.
5. Find Mean for the following Frequency Distribution by direct and Deviation methods :

X	19	21	23	25	27	29	31
f	13	15	16	18	16	15	13

6. If $n = 9$, $\sum x = 2$, $\sum y = 7$, $\sum x^2 = 134$, $\sum y^2 = 165$, $\sum xy = -129$. Find Y.
7. Write the difference between Correlation and Regression Analysis.

SECTION – C

Answer any four of the following:

8.
(a) Explain Limitations of Statistics.

(b) The data given below related to the Height and Weight of 20 persons. Construct a Bi-variate frequency table with class interval for height as 115-125, 125-135, ….. and weight as 62-64, 64-66….. write down the marginal distribution of X and Y.

Sl.no	Height (cms)	Weight (kgs)
1	170	70
2	135	65
3	136	65
4	137	64
5	148	69
6	121	63
7	117	65
8	128	70
9	148	71
10	129	62

Sl .no	Height (cms)	Weight (kgs)
11	163	70
12	139	67
13	122	63
14	134	68
15	140	67
16	132	69
17	120	65
18	148	68
19	129	67
20	152	67

5+5=10

9.

(a) Explain Component and Multiple diagram with examples.

(b) Draw Histogram frequency polygon from the following frequency distribution.

C.I. :	10-20	20-30	30-40	40-50	50-60
Frequency :	2	5	12	7	4

6+4=10

10.

(a) Define Mean, Median, Mode with example.

(b) Find the Mean and Median for the following data by Direct Method:

17,32,35,33,15,21,41,32,11,18,20,22,22,11. 6+4=10

11.

(a) If Quartile deviation = 17, $Q_1 = 22$ find Q_3.

(b) Calculate the Spearman's rank correlation coefficient for the following data:

X	35	39	38	42	44	46	51	54	55	56
y	40	32	39	42	41	31	50	52	46	55

4+6=10

12.

(a) What are the applications of a trend?

(b) Fit a trend line to the following data by the method of the given several averages:

X(years):	2009	2010	2011	2012	2013	2014
Y(Sales):	62	58	54	64	56	55

4+6=10

SEP 2016 NUMERICAL METHOD 80 MARKS

SECTION –A 10*2=20

I. Answer any ten of the following questions:

1. Write the number 0.004854 in normalized floating point mode.
2. Add 0.4546E5 to 02439E5.
3. Write the difference between algebraic and transcendental equations.
4. What is direct method?
5. Write the formula to find the root of an equation using secant method.
6. Define eigen value.
7. What is interpolation?
8. Write the Newton Gregory forward interpolation formula for the equal intervals.
9. Define Shift operator 'E'.
10. What is numerical Differentiation?
11. What is an upper triangular matrix?
12. Write the Simpaon's $\binom{3}{8}^{\text{th}}$ rule formula for numerical integration.

SECTION – B

II. Answer any six of the following questions:

13. Explain representation of a binary number in normalized floating point mode.
14. Find the root of an equation $x^4 - x - 10 = 10$ that lies between 1.8 and 2.0 using Bisection method carry out 3 iterations.
15. Find the root of an equation $x^3 - 5x + 3 = 0$ that lies near $x = 0$ using Newton Raphson method carry out 4 iterations.
16. Solve using Gauss elimination method.

$$x + 2y + z = 3$$
$$3x + 3y + 3z = 10$$
$$3x - y + 2z = 13$$

17. Explain Gauss Jordan method.
18. Fit a straight line $y = a + bx$ for the following data using least square method.

x:	1	3	4	6	8	9	11	14
y:	1	2	4	4	5	7	8	9

19. Using Lagrange's method find the value of y when $x = 10$ from the following data:

x:	5	6	9	11
y:	12	13	14	16

20. Evaluate $I = \int_0^1 \frac{dx}{1+x^2}$ dividing the interval [0,1] into six equal parts by using trapezoidal rule.

SECTION – C

III. Answer any three of the following:

21.
   (a) Using Regula Falsi method find a real root of an equation $x^3 - 2x - 5 = 0$, corrected to 3 decimal places.
   (b) Explain Secant method to find the root of an equation.
22.

(a) Using Gauss Siedal method solve the following system of equations, using the initial approximation (0,0,0) carry out 3 iterations.

$$20x + y - 2z = 17$$
$$3x + 20y - z =- 18$$
$$2x - 3y20z = 25$$

(b) Find the eigen values and eigen vectors of the matrix $A = \begin{bmatrix} 5 & 4 \\ 1 & 2 \end{bmatrix}$.

23. Using LU decomposition method solve the following system of equations.

$$x + y + z = 1$$
$$3x + y - 3z = 5$$
$$x - 2y - 5z = 10$$

24.

(a) From the following table compute $\frac{dy}{dx}$ $at\ x = 1$.

x:	1	2	3	4	5	6
y:	1	8	21	64	125	216

(b) Using suitable interpolation formula find $f(38)$.

x:	40	50	60	70	80	90
$y = f(x)$:	184	204	226	250	276	304

25.

(a) Using the following data fit a straight line $y = ax + b$ using least square method.

x:	5	10	15	20	25
y:	16	19	23	26	30

(b) Evaluate $I = \int_0^1 \frac{x}{1+x^2} dx$ using Simpson's $\binom{1}{3}^{\text{rd}}$ rule by taking 6 equal parts.

SEP 2025 NUMERICAL AND STATISTICAL METHOD 80 Marks

SECTION –A 10*2=20

Answer any ten of the following questions:

1. Define transcendental equation. Give an Example.
2. Define eigen value and eigen vector.
3. Write a formula to find rot of an equation using Seccant-Method.
4. For which system of equation Gauss –Seidel method is applicable.
5. What is an upper triangular matrix?
6. List the applications of statistics in business.
7. State the general rules for constructing diagrams and graphs.
8. Explain any 4 parts of table.
9. Define arithmetic mean and also write down the types of mean.
10. Define co-efficient of Variation.
11. Define ogives or cumulative frequency curves.
12. Define Probability with examples.

SECTION – B

Answer any six of the following questions: 4*5=20

13. By using Bisection method find the root of an equation $x^3 - 4x - 9 = 0$. What lies between 2 and 3, carry out 3 iterations.
14. Solve the Gauss elimination method. $2x + y + z = 0, \quad 3x + 2y + 3z = 18, \quad x + 4y + 9z = 16.$
15. Distinguish between primary data and secondary data and also discuss various method of collecting primary data.
16. Calculate the median for the following data:

Profits (RS .in lakhs)	30-32	32-34	34-36	36-38	38-40	40-42	42-44
No.of camp	02	09	25	30	49	62	39

17. If you Pull 2 cards out of a deck, what is the probability that both are spades?

SECTION – B

Answer any six of the following questions: 4*10=40

18. A) Explain the Newton – Ruphson Method to find roots of an equation.
    B) Find the root of an equation $x - e^x = cosx$ by using Regula-falsi method.
19. A) Using LU –decomposition Method. Solve the following System of equation.
    $$x + y + z = 1, \quad 3x + y - 3z = 5, \quad x - 2y - 5z = 10$$
    B) Determine the largest eigen value of matrix.
    $$\begin{bmatrix} 1 & 6 & 1 \\ 1 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$
20. A) For the following frequency distribution of height of students, draw Histogram.

Heights (in cms) :	140-145	145-150	150-155	155-160	160-165	165-170
No.of Students :	4	10	18	20	19	6

    B) Explain the Pictographs and Cartographs.

21. A) Calculate the Quartile deviation and its co-efficient for the following frequency distribution.

Daily wages (in Rs) :	3	5	7	9	11	13	15	17	19
No.of Persons :	2	10	12	15	20	13	12	10	4

    B) Define Mean, Median & Mode with examples.

22. A) Three coins are tossed at atime, Find

i) Probability of getting 2 heads.

ii) Probability of getting no heads.

iii) Probability of getting 2 heads and 1 tail.

B) Write down the probability function, range, mean, variance of Bernoulli distribution if [P=0.2].